

Truong Hoai An Luu

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SUMMARY

Senior Electrical Engineering student at Iowa State University graduating May 2027, seeking full-time opportunities in power or VLSI.

EDUCATION

Iowa State University (Ames, IA) Graduating May 2027
Bachelor of Science in Electrical Engineering
FE Electrical & Computer Exam: Scheduled December 29, 2026
Des Moines Area Community College (Des Moines, IA) Graduated May 2024
Associate in Science

WORK EXPERIENCE

Quality Engineering Intern June 2026 – August 2026
Electrical Power Products (EP2) (Des Moines, IA)

- Developed VKS work instructions for electrical panel testing and assembly, translating hands-on processes into standardized step-by-step procedures.
- Shadowed experienced testers and performed supervised functional testing using Omicron, SEL/GE protection relays, AMETEK UPLC, multimeters, and AC/DC power supplies.
- Advanced the 8-inch PSE&G Panel Assembly WI through Level 3 review and Function Testing WI through most of Level 2, coordinating reviews across Engineering, Testing, QC, Quality Assurance, manufacturing, and leadership.
- Supported a company-wide inventory audit using SQL-based systems and Excel to verify inventory, orders, panel status, and manufacturing records.

PROJECTS

Eddy Current (EC) Probes for Conductivity Measurements *(In Progress)* August 2026 – May 2027

- Collaborating in a six-member engineering team to design a portable eddy-current-based system, developing project requirements and evaluating design concepts based on client and user needs.
- Applying the engineering design process through requirements definition, stakeholder communication, concept generation, and technical evaluation toward prototype development and testing.

Multifunction Programmable Integrated Circuit | *Cadence Virtuoso* April 2026 – May 2026

- Designed and simulated a four-function programmable circuit integrating non-inverting/inverting amplifiers, a 16-tap digital potentiometer, and DAC with 2-bit mode selection.
- Designed CMOS circuitry including NMOS/PMOS bidirectional switches, 2-to-4/4-to-16 decoders, resistor-string networks, and programmable amplifier stages.
- Completed schematic simulation, physical layout, DRC, and LVS verification of the integrated circuit.

Power Electronics System Simulation | *PLECS* November 2025 – December 2025

- Implemented and simulated a PLECS power-electronics system with boost and buck converters, transformer interfaces, and multiple electrical loads.
- Analyzed DC-bus regulation, three-phase voltages/currents, converter waveforms, and average power to evaluate system performance.

COURSEWORK

Power System Analysis, Economic Systems for Electric Power Planning, Energy Systems & Power Electronics, Integrated Electronic & Circuit Systems, Microelectronics Fabrication

SKILLS

Technical: Power Electronics, CMOS/VLSI Design, Circuit Design & Simulation, Electrical Testing & Troubleshooting
Software: PLECS, Cadence Virtuoso, MATLAB/Simulink, LTspice, Quartus, QuestaSim, VKS, Excel
Equipment: Omicron, Oscilloscope, Multimeter, Function Generator, DC Power Supply
Programming: C/C++, Python, MATLAB, SQL, Verilog, VHDL